

Version Control Systems

An Overview of Google Docs

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Algorithm Basics and BW

Recap: What is Version Control?

From last class:

- ▶ Tracking changes to documents over time
- ▶ Seeing who changed what and when
- ▶ Ability to restore previous versions
- ▶ Recovery from mistakes

 **GitHub approach:** Explicit commits with messages

Today's approach: Automatic tracking in  Google Docs

Why Google Docs for Version Control?

- ▶ You probably already use it!
- ▶ Version control built-in (automatic)
- ▶ Great for collaborative documents
- ▶ No commit messages needed
- ▶ Real-time collaboration
- ▶ Familiar interface

Key difference from GitHub:

- ▶  **GitHub:** You decide when to save versions (commits)
- ▶  **Google Docs:** Automatically saves every change

When to Use Google Docs Version Control

Best for:

- ▶ Collaborative writing (reports, proposals, meeting notes)
- ▶ Documents with frequent small changes
- ▶ Teams that need real-time editing
- ▶ Projects where you want automatic backups

Less ideal for:

- ▶  Code or technical files
- ▶ Projects requiring explicit version labels
- ▶  Large datasets or structured data

Today's Hands-On Project

Scenario: Creating a team meeting agenda that evolves over time

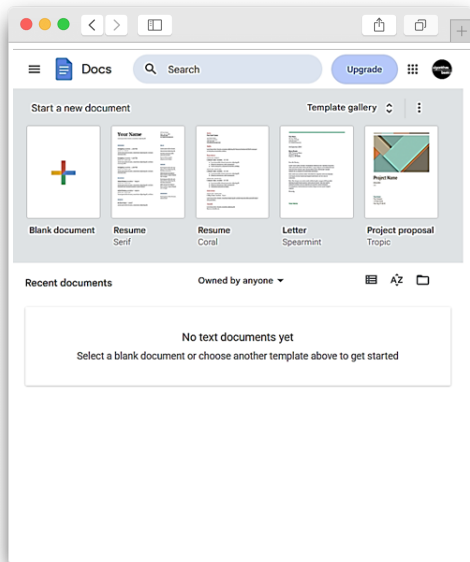
What we'll do:

1. Create a new  Google Doc
2. Build a meeting agenda with multiple updates
3. Use version history to track changes
4. Name specific versions
5. Restore a previous version
6. See who made what changes (if collaborating)

Step 1: Create a New Google Doc

1. Go to <https://docs.google.com>
2. Sign in with your **G** Google account (or university account)
3. Click “Blank” to create new document
4. Click “Untitled document” at top left
5. Rename to:
Team Meeting Agenda - Version Control Demo

Note:  Google Docs starts tracking versions immediately!



Step 2: Create Initial Agenda

Type the following content:

Meeting Agenda

Team Meeting - January 27, 2026

Time: 2:00 PM - 3:00 PM

Location: Conference Room A

Attendees:

- Sarah Johnson (Project Manager)
- Mike Chen (Developer)

Agenda Items:

1. Project status update
2. Budget review

Google Docs is automatically saving and tracking this!

Step 2: Create Initial Agenda

Team Meeting Agenda - Version Control Demo

File Edit View Insert Format Tools Extensions Help

100% | Normal text | Times ... | 14 | B I U A |

Meeting Agenda

Team Meeting - January 27, 2026
Time: 2:00 PM - 3:00 PM
Location: Conference Room A

Attendees:

- ☐ Sarah Johnson (Project Manager)
- ☐ Mike Chen (Developer)

Agenda Items:

1. Project status update
2. Budget review

Step 3: Make Changes Over Time

Let's simulate making updates as the meeting approaches:

Update 1 - Add more agenda items (wait 1 minute, then add):

▶ Add to Agenda Items section:

3. Timeline discussion
4. Risk assessment
5. Next steps

Update 2 - Add more attendees (wait 1 minute, then add):

▶ Add to Attendees:

- Jennifer Lee (Designer)
- Robert Martinez (QA Lead)

Step 4: More Realistic Changes

Update 3 - Modify an agenda item:

- ▶ Change “Budget review” to “Q1 Budget Review - \$50K allocation”

Update 4 - Add meeting notes:

- ▶ At the end of document, add:

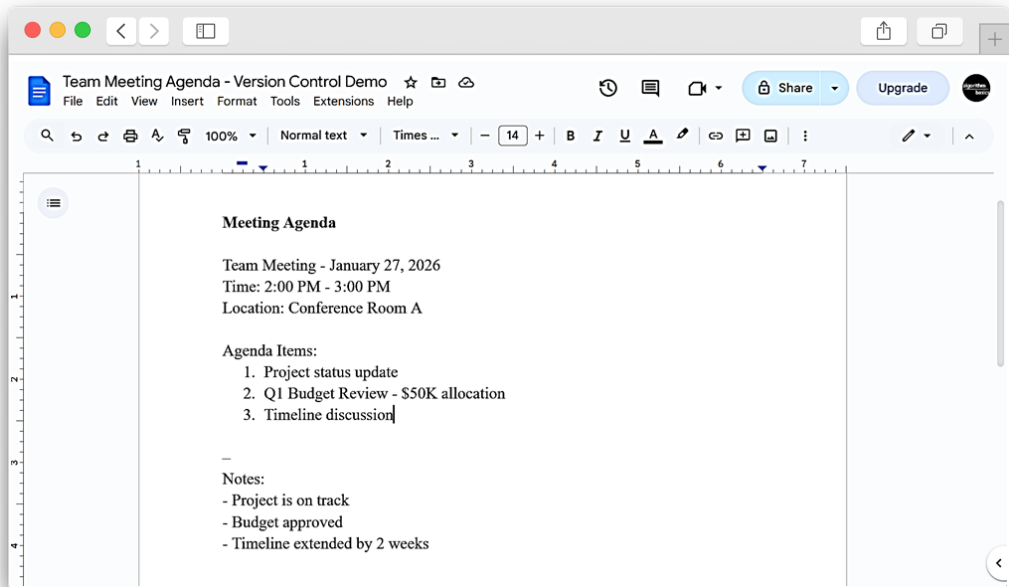
Notes:

- ▶ Project is on track
- ▶ Budget approved
- ▶ Timeline extended by 2 weeks

Update 5 - Simulate a mistake:

- ▶ Delete the entire “Attendees” section
- ▶ Delete agenda items 4 and 5

Step 4: More Realistic Changes



The screenshot shows a web-based document editor interface. At the top, there's a title bar with window controls (red, yellow, green buttons) and navigation arrows. Below the title bar, the document title is "Team Meeting Agenda - Version Control Demo". The menu bar includes "File", "Edit", "View", "Insert", "Format", "Tools", "Extensions", and "Help". On the right side of the menu bar, there are "Share" and "Upgrade" buttons. Below the menu bar is a toolbar with various editing tools like search, undo, redo, bold, italic, underline, text color, background color, and list creation. The main editing area contains the following text:

Meeting Agenda

Team Meeting - January 27, 2026
Time: 2:00 PM - 3:00 PM
Location: Conference Room A

Agenda Items:

1. Project status update
2. Q1 Budget Review - \$50K allocation
3. Timeline discussion

Notes:

- Project is on track
- Budget approved
- Timeline extended by 2 weeks

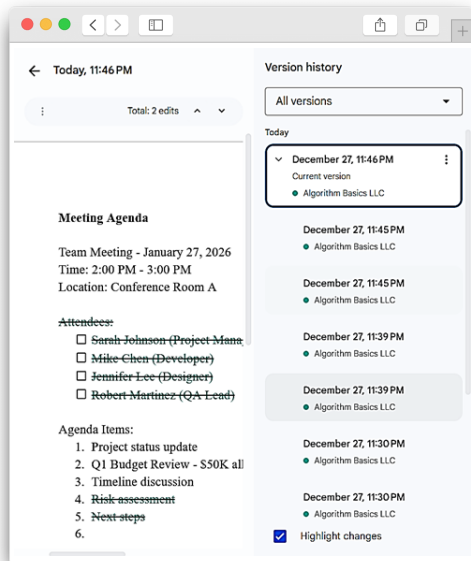
Step 5: Access Version History

Now let's see all our changes:

1. Click "File" in the menu
2. Click "Version history"
3. Click "See version history"
4. Right side panel shows all versions

What you'll see:

- ▶ Timeline of changes
- ▶ Color-coded changes
- ▶ "Today" as the grouping
- ▶ Automatic timestamps



Understanding the Version History Panel

Left side: Your current document

Right side panel shows:

- ▶ Timestamp of each change
- ▶ Your name (who made the change)
- ▶ Colored indicators showing what changed
- ▶ Option to name versions
- ▶ Option to restore versions

Click on different timestamps to see your document at different points in time

Step 6: Explore Changes

Click through your version history:

1. Click on the earliest version (when you first created it)
2. Notice the document shows only your initial content
3. Click on later versions one by one
4. Watch how content appears/disappears
5. See highlighted changes:
 - ▶ Green = additions
 - ▶ ~~Strikethrough~~ = deletions

Key insight: Every keystroke is potentially recoverable!

Step 7: Name Important Versions

You can label important versions for easy reference:

1. In version history, find the version right before you deleted content
2. Click the three dots (...) next to that version
3. Click “Name this version”
4. Type:
Complete agenda before edits
5. Click  “Save”

When to name versions:

- ▶ Before major changes
- ▶ After completing a milestone
- ▶ Before sharing with stakeholders
- ▶ Final versions before submission

Step 8: Restore a Previous Version

Let's recover the content we deleted:

1. In version history, click on your named version:
Complete agenda before edits
2. Review the content - you should see attendees and all agenda items
3. Click "Restore this version" at the top
4. Click "Restore" to confirm
5. The version history panel closes
6. Your document now shows the restored content!

Important: The restore creates a NEW version. Nothing is lost.

Advanced Feature: Show Changes Mode

Toggle the “Show changes” checkbox:

1. Open version history again
2. At the top of the right panel, find “Show changes” checkbox
3. When checked: Shows additions and deletions highlighted
4. When unchecked: Shows clean version as it appeared then

Use “Show changes” to:

- ▶ Review what was edited
- ▶ Understand evolution of document
- ▶ Find when specific content was added/removed

Step 9: Make More Changes

Let's see version history in action:

1. Close version history (click X or "Back to document")
2. Add a new section at the end:

Action Items

1. Sarah - Send updated timeline to stakeholders
2. Mike - Complete API integration by Feb 1
3. Jennifer - Finalize design mockups

3. Open version history again
4. You'll see your new changes at the top!

Collaboration Features

If you're working with a teammate:

1. Click "Share" button (top right)
2. Add your teammate's email
3. Give them "Editor" access
4. Have your teammate make some changes
5. Open version history
6. You'll see their name next to their changes!

In real projects: This shows exactly who contributed what

Practice Exercise

Create a new document for practice:

1. Create a project proposal document
2. Add: Title, Executive Summary, Budget, Timeline
3. Make at least 5 separate edits (wait 30 seconds between each)
4. Name at least 2 important versions
5. Delete some content, then restore it using version history
6. Explore the detailed revision history

Challenge: Check how granular you can get with version history

Google Docs vs GitHub: Key Differences

Feature	 Google Docs	 GitHub
Version creation	Automatic	Manual (commits)
Best for	Documents, writing	Code, data, files
Version messages	No	Yes (required)
Collaboration	Real-time	Asynchronous
Learning curve	Low	Medium
Control	Less explicit	More explicit

Both achieve the same goal: tracking changes over time!

Key Takeaways: Google Docs Version Control

- ▶ Version control is **automatic** - no action needed
- ▶ Access via File → Version history
- ▶ Every change is tracked with timestamp
- ▶ Green highlights = additions, strikethrough = deletions
- ▶ Name important versions for easy reference
- ▶ Restoring a version creates a new version (nothing lost)
- ▶ Perfect for collaborative document work
- ▶ Shows who made what changes

Questions and Discussion